

Computational Discovery of a Novel Yttrium Silicide as an Ambient Pressure Superconductor

Shuo Tao, Ying Sun, and Li Zhu*

Metal silicides have garnered significant attention due to their diverse crystalline structures, tunable electronic properties, and potential applications in superconducting devices. While numerous rare-earth metal silicides (REMSs) demonstrate superconductivity at ambient pressure, the known yttrium silicide phases exhibit extremely low superconducting critical temperatures ($T_c < 1$ K), limiting their practical applications. Through comprehensive crystal structure prediction and first-principles calculations, the discovery of a novel $I4_1/amd$ -YSi₆ phase that exhibits superconductivity with a T_c of 8.4 K at ambient pressure is reported. This is significantly higher than any previously reported binary yttrium silicide. This electron-phonon coupling calculations reveal that the unique dumbbell silicon network in this structure plays a crucial role in mediating strong electron-phonon interactions, with a coupling parameter $\lambda = 0.69$. This finding expands the fundamental understanding of structure-property relationships in metal silicides and highlights the potential of yttrium silicides as promising candidates for next-generation superconducting electronics, including quantum circuits and superconducting qubits.

1. Introduction

Superconducting materials that operate at accessible temperatures and ambient pressure remain highly sought after for technological applications ranging from quantum computing to magnetic resonance imaging.^[1,2] Silicon-based superconductors are particularly attractive for their potential compatibility with established semiconductor fabrication technologies, enabling seamless integration into electronic devices.^[3,4] Therefore, metal silicides have emerged as promising candidates due to their diverse structural motifs, tunable electronic properties, and stability under various conditions.^[5,6]

The exploration of superconductivity in silicide-based materials has followed several distinct pathways. Type-I clathrate structures (M₈Si₄₆) represent one important class of silicon-based superconductors. A breakthrough in this area occurred when

(Na,Ba)_xSi₄₆ was found to exhibit type-II superconductivity with a transition temperature (T_c) of approximately 4 K.^[7] This discovery demonstrated the first superconductor consisting of a silicon sp³ covalent network, highlighting the potential of silicon frameworks to mediate superconductivity.^[7–9] Various doping strategies have since been explored to enhance superconducting properties in silicon-rich systems. Heavily boron-doped silicon carbide (SiC), for instance, behaves as a type-I superconductor, sharing characteristics with other boron-doped semiconductor superconductors.^[3,10,11]

Transition metal silicides constitute another important category, where the metallic-type bonding introduced by transition elements significantly influences superconducting behavior. The substitution of α -ThSi₂ with Rh or Ir substantially increases the T_c of the resulting

compounds ThRh_{0.96}Si_{1.04} and ThIrSi, reaching the values of 6.45 and 6.5 K, respectively.^[12] This pronounced enhancement in superconductivity stems from changes in the electronic structure due to the introduction of transition metals with partially filled d-orbitals. Recent advances in thin-film fabrication techniques have further expanded the potential of superconducting silicides. The A15 phase of vanadium silicide (V₃Si) can achieve remarkable T_c values exceeding 13 K when fabricated by precisely controlling the interdiffusion between metallic vanadium films and silicon substrates.^[1] Similarly, tungsten silicide W₅Si₃ exhibits bulk superconductivity with $T_c \approx 2.7$ K and an upper critical field of approximately 5200 Oe, confirming its classification as a type-II superconductor.^[13] The exploration of compositional complexity in silicides has also yielded promising results. High-entropy silicides with W₅Si₃-type structure demonstrate T_c values of 3.2–3.3 K, suggesting that increased configurational entropy may enhance superconducting properties.^[14]

The investigation of ternary silicides, particularly those adopting the AlB₂-type structure, has revealed additional avenues for enhancing superconductivity. YbGa_{1.1}Si_{0.9}, which crystallizes in the AlB₂-type structure with disordered Ga and Si atoms forming honeycomb layers, functions as a type-II superconductor with $T_c = 2.4$ K.^[15] More impressively, Ca(Al_{0.5}, Si_{0.5})₂, also adopting the AlB₂-type structure, exhibits a substantially higher T_c of 7.7 K, demonstrating the potential of this structural motif for superconducting applications.^[16]

S. Tao, Y. Sun, L. Zhu
Department of Physics
Rutgers University
Newark, NJ 07102, USA
E-mail: li.zhu@rutgers.edu

© 2025 The Author(s). Annalen der Physik published by Wiley-VCH GmbH. This is an open access article under the terms of the [Creative Commons Attribution](#) License, which permits use, distribution and reproduction in any medium, provided the original work is properly cited.

DOI: 10.1002/andp.202500210

Transitioning from these silicon-based superconductors, rare-earth metal silicides (REMSs)^[17] have attracted particular attention due to their exceptional properties relevant to electronic and superconducting applications. Beyond their superconducting capabilities, REMSs offer low Schottky barrier heights (SBH),^[18] enabling the formation of non-rectifying electrical junctions that allow efficient charge carrier flow without rectification. This attribute is crucial for minimizing energy losses in electronic devices and has positioned REMSs as promising candidates for advanced applications in quantum technologies. Specifically, metal silicides show great potential for implementation in Josephson junctions (JJs), which are fundamental components of superconducting quantum interference devices (SQUIDs) and quantum computing qubits.^[2] The precise control of supercurrents with minimal resistance that REMSs can provide is essential for enhancing the efficiency, stability, and scalability of these quantum devices.

A significant advantage of rare-earth metal silicides (REMSs) is their compatibility with silicon-based fabrication processes, facilitating the integration of Josephson junctions (JJs) into existing silicon technology platforms. Studies have demonstrated that REMSs can form low-resistance, stable contacts with silicon, which is advantageous for integrating superconducting devices into existing silicon technology platforms.^[4] The ability to grow high-quality REMS films epitaxially on silicon substrates using established techniques, such as rapid electron beam heating and lamp heating, enables the seamless incorporation of these materials into semiconductor manufacturing processes. Research has shown that rapid annealing techniques can form thin epitaxial films of REMSs on silicon substrates, which is beneficial for device fabrication.^[19] This compatibility is crucial for scaling up superconducting electronic devices and quantum computing technologies, as it allows for the integration of superconducting components with existing silicon-based infrastructure. The seamless incorporation of REMSs into semiconductor manufacturing processes facilitates the development of advanced electronic and quantum devices.

Among the rare-earth elements, yttrium stands out for both its practical advantages and scientific interest. As the fourth most abundant rare-earth element in the Earth crust, yttrium offers greater accessibility and cost-effectiveness for large-scale applications compared to other rare-earth metals. Chemically, yttrium shares many properties with lanthanide elements, forming stable silicides with diverse structural configurations. Complementing the versatility of yttrium, silicon is renowned for forming a wide array of stable and metastable phases, many of which exhibit intriguing properties. Numerous studies have demonstrated superconductivity in various silicon allotropes^[3,7–9,20–25] and compounds.^[6–9,23–25]

Despite the promise of yttrium silicides for superconducting applications, a significant limitation has persisted: among the currently known stable phases, only *Cmcm*-YSi and *P6₃/mcm*-Y₅Si₃ exhibit superconductivity at ambient pressure, and their critical temperatures remain extremely low ($T_c < 1$ K).^[26] This deficiency severely restricts their practical utility for technological applications that require accessible operating temperatures. The identification of yttrium silicide structures with substantially higher T_c values would represent a significant advancement in

the field and could enable new applications in superconducting technologies.

In this work, we address this challenge through computational structure prediction and first-principles calculations. We performed extensive structure searching on the Y-Si binary system at elevated pressures (10 GPa) using the particle swarm optimization (PSO) algorithm to identify novel phases with enhanced stability and superconducting properties. Our investigation reveals a thermodynamically stable *I4₁/amd*-YSi₆ phase at 10 GPa that maintains dynamical stability when decompressed to ambient pressure. This structure exhibits metallic behavior and superconductivity, with a T_c of 8.4 K at ambient pressure—substantially higher than any previously reported binary yttrium silicide.

2. Computational Methods

We performed comprehensive structure searching across the Y-Si binary system using the CALYPSO (Crystal structure AnaLYsis by Particle Swarm Optimization) code.^[27,28] The search encompassed a wide range of stoichiometries, including Y₂Si, YSi, YSi₂, YSi₄, YSi₆, YSi₈, and YSi₁₀, with 2–4 formula units per unit cell. Structure searches were conducted at a pressure of 10 GPa to identify phases that may be stabilized under compression.

For the structure optimization during the CALYPSO prediction process, we employed our in-house Reformpy code, which accelerates structural optimization by mapping configurations to a fingerprint space that captures essential structural features.^[29] This technique biases the search toward high-symmetry, low-energy configurations by identifying and preserving structural motifs that correlate with energetic stability, enabling efficient navigation of the high-dimensional potential energy surface (PES) within each CALYPSO generation. In this work, the Reformpy code is interfaced with the Vienna Ab initio Simulation Package (VASP).^[30] We employed the Perdew–Burke–Ernzerhof (PBE) generalized gradient approximation^[31,32] for the exchange–correlation functional. The frozen-core all-electron projector augmented wave (PAW) method^[33] was used with potentials treating 4s²4p⁶4d¹5s² and 3s²3p² as valence electrons for Y and Si, respectively. Electronic wave functions were expanded in a plane-wave basis set with a cutoff energy of 500 eV, which was determined through convergence testing to provide energy differences accurate to within 1 meV/atom. The Brillouin zone was sampled using Monkhorst–Pack k-point meshes^[34] with a uniform grid spacing of $0.04 \times 2\pi \text{ \AA}^{-1}$, ensuring sufficient sampling density for accurate total energy calculations.

Electronic band structures and density of states (DOS) were calculated using the local modified Becke–Johnson (LMBJ) exchange potential,^[35,36] which provides more accurate band gaps and electronic structures compared to standard PBE calculations. The total and orbital-projected DOS were computed with a denser k-mesh (1.5 times the density used for structural optimization) to capture the fine details of the electronic structure near the Fermi level.

The dynamical stability of the predicted *I4₁/amd*-YSi₆ phase was assessed through phonon calculations using the finite displacement method as implemented in the PHONOPY code.^[37,38] For these calculations, we constructed supercells with dimensions of at least $3 \times 3 \times 3$ primitive cells and applied displacements of 0.01 Å to generate the force constants. The phonon

dispersion and density of states were calculated along high-symmetry paths in the Brillouin zone, and the absence of imaginary frequencies was used as the criterion for dynamical stability. To investigate the superconducting properties of the dynamically stable $I4_1/amd$ -YSi₆ phase, we conducted electron-phonon coupling (EPC) calculations within the framework of density functional perturbation theory (DFPT) using the Quantum ESPRESSO package.^[39] We employed ultrasoft pseudopotentials from the Standard Solid-State Pseudopotentials (SSSP) library,^[40] with kinetic energy cutoffs of 50 Ry for wavefunctions and 400 Ry for charge density, which were determined through convergence testing of the phonon frequencies. The electron-phonon coupling calculations were performed using a $5 \times 5 \times 5$ q-point mesh for phonons and a $10 \times 10 \times 10$ k-point mesh for electrons. The q-point mesh density was systematically tested for convergence (see [Supporting Information](#)). Electronic structure calculations employed Methfessel–Paxton smearing with a width of 0.010 Ry. For the electron-phonon calculations, we used Gaussian broadening for the delta functions, systematically varying the width from 0.005 to 0.050 Ry in steps of 0.005 Ry. For each q-mesh density, we performed calculations at multiple broadening values and extrapolated to the zero-broadening limit to obtain converged values of λ and T_c . The self-consistency threshold for the lattice dynamics calculations was set to 1×10^{-14} Ry to ensure high precision in the computed phonon properties. The superconducting critical temperature (T_c) was estimated using the Allen–Dynes-modified McMillan equation:^[41,42]

$$T_c = \frac{\omega_{\log}}{1.2} \exp \left[-\frac{1.04(1 + \lambda)}{\lambda - \mu^*(1 + 0.62\lambda)} \right] \quad (1)$$

where λ is the electron-phonon coupling parameter, μ^* is the Coulomb pseudopotential (set to 0.1, a typical value for metals), and $\omega_{\log}$ is the logarithmic average frequency defined as:

$$\omega_{\log} = \exp \left[\frac{2}{\lambda} \int \frac{d\omega}{\omega} \alpha^2 F(\omega) \log \omega \right] \quad (2)$$

The Eliashberg spectral function $\alpha^2 F(\omega)$ and the EPC parameter λ were calculated through direct numerical integration of the phonon linewidths over the Brillouin zone. The partial EPC parameter $\lambda(\omega)$ was evaluated as:

$$\lambda(\omega) = 2 \int_0^\omega \frac{\alpha^2 F(\omega')}{\omega'} d\omega' \quad (3)$$

This approach allows for the identification of the phonon modes that contribute most significantly to the electron-phonon coupling and, consequently, to the superconducting properties of the material.

3. Results and Discussion

3.1. Thermodynamic Stability and Phase Relationships

To identify thermodynamically viable yttrium silicide phases, we evaluated the formation enthalpies of all structures discovered during our crystal structure prediction searches. For each com-

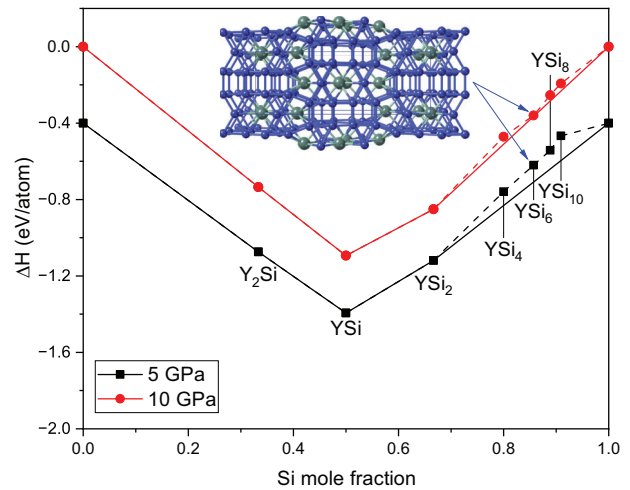


Figure 1. Formation enthalpy of Y–Si binary system at 5 GPa (black) and 10 GPa (red) with respect to silicon mole fraction. Dashed lines connect data points, and solid lines denote the convex hull. The 5 GPa plot is shifted by -0.4 eV relative to the 10 GPa plot for clarity. Insets show top-down views of $I4_1/amd$ -YSi₆ (along b-axis) with dark green and blue spheres representing Y and Si atoms, respectively.

position Y_xSi_y , the formation enthalpy per atom was calculated as:

$$\Delta H(Y_xSi_y) = [H(Y_xSi_y) - xH(Y) - yH(Si)] / (x + y) \quad (4)$$

where $H(Y_xSi_y)$ is the enthalpy of the compound, and $H(Y)$ and $H(Si)$ are the enthalpies of elemental yttrium and silicon in their ground-state structures at the corresponding pressure.

Figure 1 presents the formation enthalpies of various Y–Si compositions at 5 and 10 GPa, with the convex hull indicated by solid lines. Points lying on the convex hull represent thermodynamically stable phases, while those above the hull are metastable with respect to decomposition into either other Y–Si compounds or the elemental solids. We find Y_2Si , YSi , and YSi_2 are stable on the convex hull, which is consistent with previous studies.^[26] In addition, our analysis reveals a thermodynamically stable $I4_1/amd$ -YSi₆ phase at 10 GPa. This silicon-rich composition represents a previously unreported stoichiometry in the Y–Si system. The dynamical stability of the $I4_1/amd$ -YSi₆ phase was assessed through phonon dispersion calculations at various pressures. The absence of imaginary phonon frequencies throughout the Brillouin zone confirms that this phase is dynamically stable across the entire studied pressure range (0–10 GPa), making it potentially synthesizable and recoverable at ambient pressure. This pressure-independent stability has important implications for potential synthesis strategies, suggesting that the $I4_1/amd$ -YSi₆ phase could be formed at high pressure and successfully recovered to ambient conditions.

3.2. Structural Characteristics

The $I4_1/amd$ -YSi₆ structure exhibits distinct structural features that are critical to understanding its electronic and superconducting properties. **Figure 2** illustrates the key structural ele-

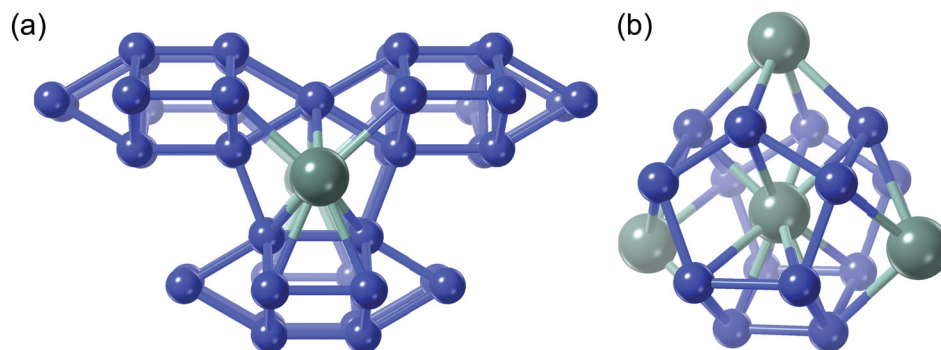


Figure 2. Crystal structures of the predicted YSi_6 phases. a) The 3D dumbbell silicon network in $I_{41}/\text{amd-YSi}_6$, highlighting the interconnected Si–Si units that form a framework with channels along crystallographic axes. b) Local atomic environment of the 10-coordinated Y atom in $I_{41}/\text{amd-YSi}_6$, showing the distorted polyhedron arrangement of neighboring Si atoms.

ments of this phase. The structure features an unusual 3D framework of silicon atoms arranged in dumbbell configurations. Each silicon dumbbell consists of two Si atoms with a shortened Si–Si bond length of approximately 2.32 Å at ambient pressure, compared to 2.35 Å in diamond silicon. These dumbbells are interconnected through longer Si–Si bonds (2.38–2.41 Å), forming a complex tetrahedral network that creates channels along the crystallographic axes. The yttrium atoms occupy these channels and are coordinated by 10 silicon atoms in a distorted polyhedron arrangement. This coordination environment is relatively rare in silicide structures and suggests significant charge transfer between the yttrium and silicon atoms. Our Bader charge analysis^[43] confirms this hypothesis, revealing a substantial charge transfer of 1.53e per Y atom to the silicon frameworks.

The silicon framework in $I_{41}/\text{amd-YSi}_6$ bears similarity to structures found in other silicon-rich compounds such as MSi_3 ($M = \text{Ca}, \text{Y}, \text{Lu}$),^[44] but with a higher silicon content and more complex connectivity. The dumbbell motifs create a highly interconnected and rigid network, which may contribute to the structural stability across a wide pressure range. The Y–Si bond distances range from 2.97 to 3.15 Å, indicating moderate but significant bonding interactions.

3.3. Electronic Properties

To elucidate the electronic characteristics that underpin the superconducting behavior of the $I_{41}/\text{amd-YSi}_6$ phase, we calculated its electronic band structure and density of states (DOS) using the local modified Becke–Johnson (LMBJ) exchange functional,^[35,36] which provides more accurate descriptions of band structures compared to standard DFT approaches. **Figure 3** presents the results for $I_{41}/\text{amd-YSi}_6$ at 10 and 0 GPa.

The $I_{41}/\text{amd-YSi}_6$ phase exhibits metallic behavior at both investigated pressures, as clearly indicated by multiple band crossings at the Fermi level (Figure 3). The electronic bands display substantial dispersion around the Fermi energy, suggesting high electronic mobility. Remarkably, the band structure and density of states remain essentially unchanged upon decompression from 10 GPa to ambient pressure, demonstrating the robustness of the electronic structure across this pressure range. At both pressures, the electronic states near the Fermi level primarily

originate from Y 4d orbitals and Si 3s and 3p orbitals, with minor contributions from Y 5s and 5p states. This orbital composition indicates significant hybridization between yttrium d-states and silicon s- and p-states, promoting effective charge transfer. The total DOS at the Fermi level, $N(E_F)$, is a critical parameter for superconductivity according to Bardeen–Cooper–Schrieffer (BCS) theory. Our calculations indicate that $N(E_F)$ remains essentially constant when the $I_{41}/\text{amd-YSi}_6$ phase is decompressed from 10 GPa to ambient pressure. This pressure-insensitive electronic structure, combined with the changes in phonon properties discussed below, determines the superconducting behavior and supports the conventional electron–phonon coupling mechanism for superconductivity in this material.

3.4. Phonon Properties and Superconductivity

The superconducting behavior of the $I_{41}/\text{amd-YSi}_6$ phase is fundamentally linked to its phonon properties and electron–phonon coupling characteristics.^[45,46] **Figure 4** presents the phonon dispersion curves, phonon density of states (PHDOS), Eliashberg spectral function $\alpha^2F(\omega)$, and the accumulated electron–phonon coupling parameter $\lambda(\omega)$ for this phase at 10 GPa and 0 GPa.

For $I_{41}/\text{amd-YSi}_6$ at 10 GPa (Figure 4a), the phonon spectrum extends up to approximately 500 cm^{-1} and displays well-separated acoustic and optical branches. The phonon DOS reveals that the low-frequency modes (below 200 cm^{-1}) are dominated by yttrium vibrations, while the higher-frequency modes primarily involve silicon atoms. The Eliashberg spectral function $\alpha^2F(\omega)$ shows significant contributions across the entire phonon spectrum, with particularly strong coupling in the intermediate frequency range (150–300 cm^{-1}). The integrated electron–phonon coupling parameter λ reaches a value of 0.63, with approximately 60% of the coupling strength originating from silicon-dominated phonon modes above 200 cm^{-1} .

When decompressed to ambient pressure (Figure 4b), the $I_{41}/\text{amd-YSi}_6$ phase exhibits several changes in its vibrational properties. The phonon spectrum softens slightly, with the maximum frequency decreasing to about 480 cm^{-1} , and the peaks in the PHDOS become more pronounced. In addition, the electron–phonon coupling strengthens, with λ increasing to 0.69. This enhancement is primarily due to stronger coupling with intermediate-frequency modes, as evident from the more pro-

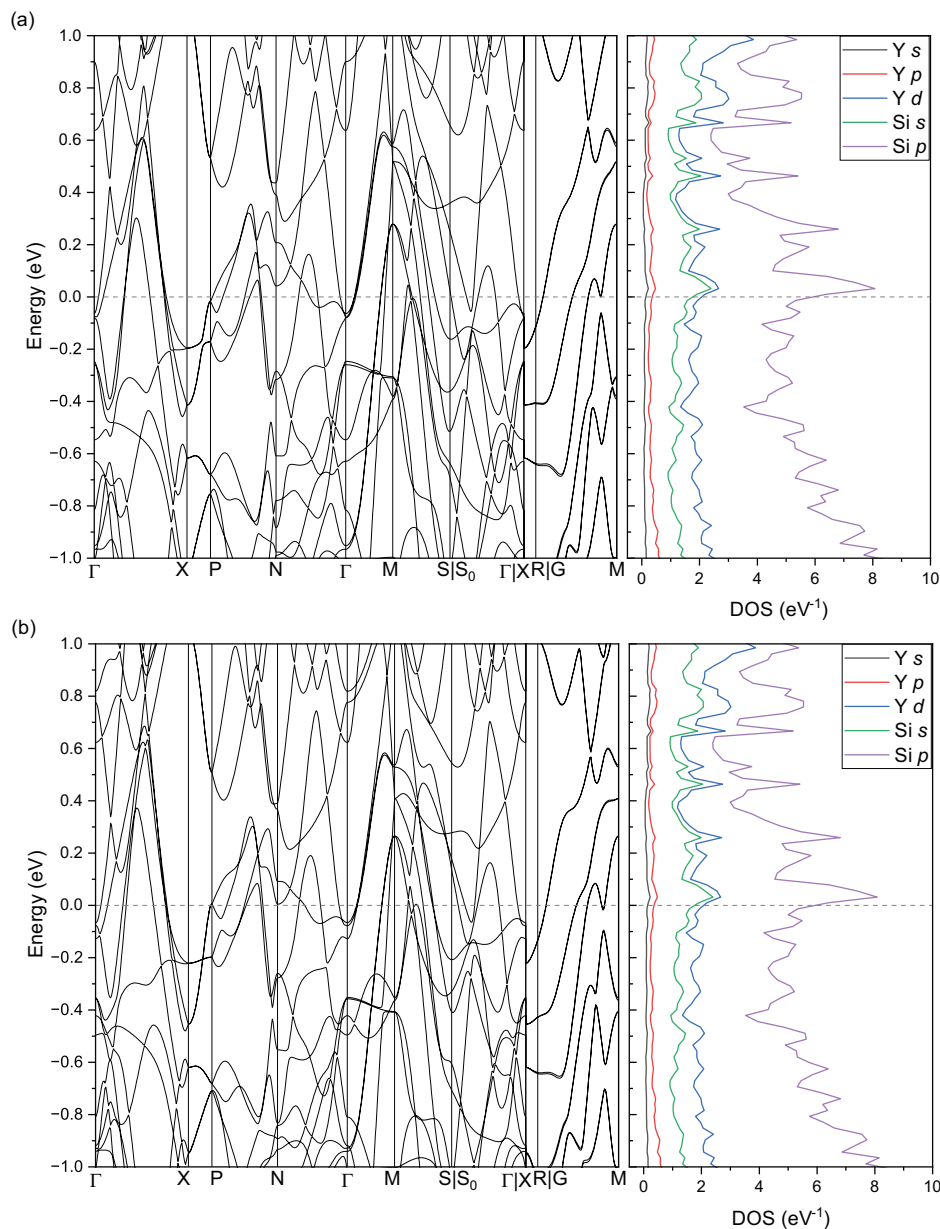


Figure 3. Electronic band structures and density of states (DOS) for the predicted YSi_6 phases. a) $I4_1/amd\text{-YSi}_6$ at 10 GPa, and b) $I4_1/amd\text{-YSi}_6$ at 0 GPa. The left panels show band structures along high-symmetry paths in the Brillouin zone, with the Fermi level (dashed line) set to zero. The right panels display the total and orbital-projected DOS, highlighting contributions from Y 4d (red), Si 3s (green), and Si 3p (blue) states. Note the significant hybridization between Y 4d and Si 3p states near the Fermi level, which is crucial for electron-phonon coupling.

nounced peaks in $\alpha^2F(\omega)$ between 150 and 250 cm^{-1} . The logarithmic average frequency ω_{log} is calculated to be approximately 347 K, which, combined with the enhanced λ , results in an estimated T_c of 8.4 K at ambient pressure using the Allen-Dynes-modified McMillan equation with $\mu^* = 0.1$.

3.5. Comparison with Known Silicide Superconductors

To contextualize our findings, we compare the predicted $I4_1/amd\text{-YSi}_6$ phase with previously reported silicide superconductors. Table 1 summarizes key parameters for selected sili-

cide superconductors, including crystal structure, superconducting transition temperature at ambient pressure.

The predicted $I4_1/amd\text{-YSi}_6$ phase with $T_c = 8.4$ K at ambient pressure represents a significant advancement over previously known binary yttrium silicides,^[26] which exhibit T_c values below 1 K. This substantial enhancement in T_c can be attributed to the unique dumbbell silicon framework that optimizes electron-phonon coupling. Among binary silicides, V_3Si stands out with a higher T_c of approximately 13 K, but this compound contains a transition metal with partially filled d shells, which often exhibit unconventional superconductivity that cannot be adequately described by conventional electron-phonon

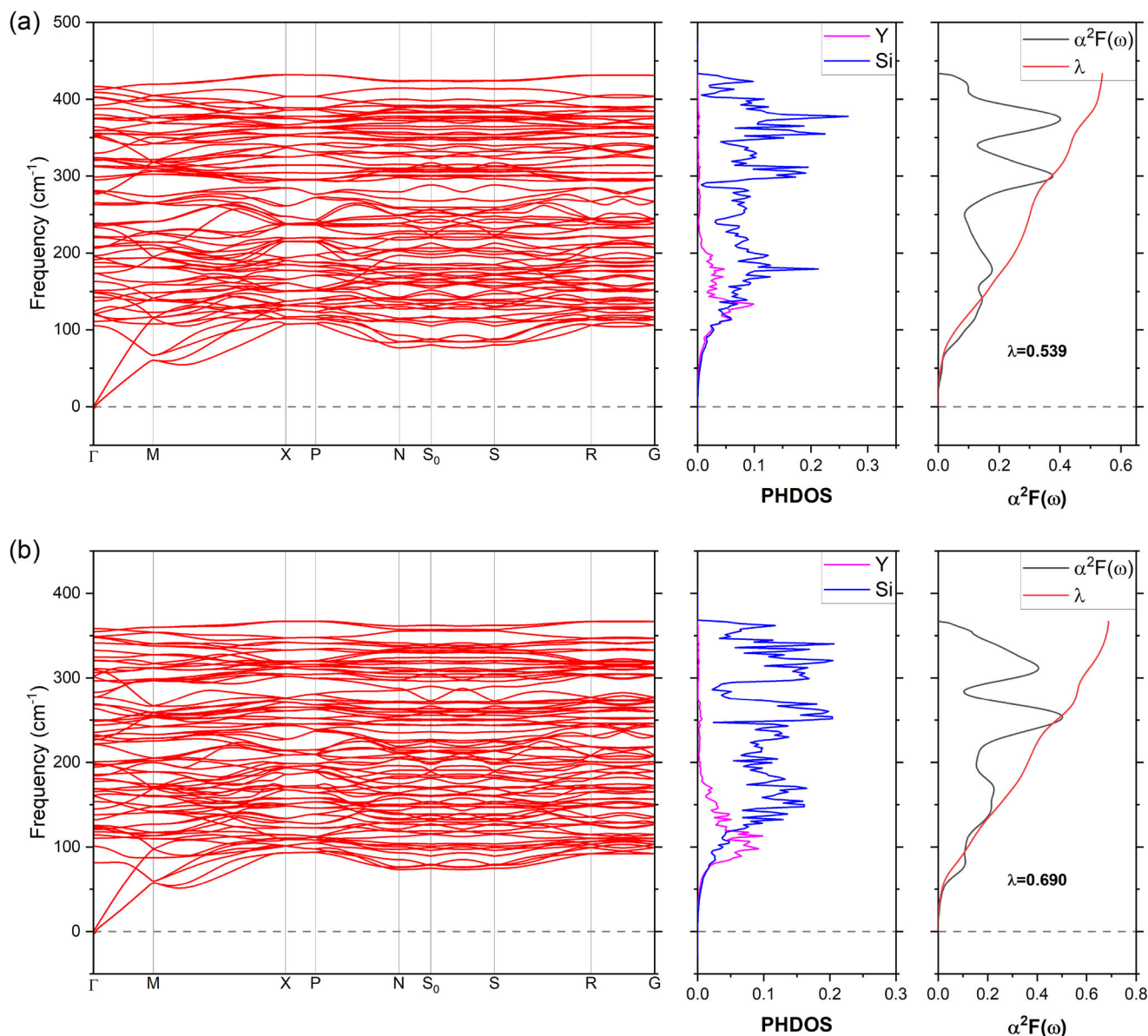


Figure 4. Phonon properties and electron-phonon coupling for YSi₆ phases. a) $I4_1/amd$ -YSi₆ at 10 GPa, and b) $I4_1/amd$ -YSi₆ at 0 GPa. For each phase, panels show (from left to right): phonon dispersion along high-symmetry paths, phonon density of states (PHDOS) with contributions from Y (pink) and Si (blue) atoms, Eliashberg spectral function $\alpha^2F(\omega)$ (black line), and accumulated electron-phonon coupling parameter $\lambda(\omega)$ (red line).

Table 1. Comparison of superconducting properties of selected silicide-based materials.

Compound	Structure	T_c [K]	λ
$I4_1/amd$ -YSi ₆	Dumbbell network	8.4	0.69
$Cmcm$ -YSi ₆ ^[26]	Orthorhombic	< 1	0.25
$P6_3/mcm$ -Y ₅ Si ₃ ^[26]	Hexagonal	< 1	0.30
(Na,Ba) _x Si ₄₆ ^[7]	Clathrate	4.0	—
Ca(Al _{0.5} ,Si _{0.5}) ₂ ^[16]	AlB ₂ -type	7.7	—
V ₃ Si ^[1]	A15	13.0	—
W ₅ Si ₃ ^[13]	Tetragonal	2.7	—

coupling theory. Such unconventional superconductors present significant challenges for computational design and prediction, as their mechanisms involve complex electronic correlations and many-body effects that remain difficult to model accurately using current first-principles methods. In contrast, our predicted $I4_1/amd$ -YSi₆ phase represents a conventional superconductor where electron-phonon coupling serves as the primary pairing mechanism, making it more amenable to computational design and discovery.

The T_c of $I4_1/amd$ -YSi₆ compares favorably with ternary silicides such as Ca(Al_{0.5},Si_{0.5})₂ ($T_c = 7.7$ K), despite the absence of additional elements that might enhance electronic properties. This suggests that the specific silicon framework in the $I4_1/amd$ -

YSi₆ phase might be particularly effective at mediating superconductivity, potentially offering a new structural motif for designing high-*T_c* silicon-based superconductors. The superconductivity in the *I*₄₁/*amd*-YSi₆ phase also differs fundamentally from that observed in clathrate silicides like (Na,Ba)_xSi₄₆, where guest atoms rattle within silicon cages. In *I*₄₁/*amd*-YSi₆, the yttrium atoms are more strongly bonded to the silicon framework, leading to different electron-phonon coupling mechanisms.

4. Conclusion

In this work, we have conducted a comprehensive computational investigation of the Y-Si binary system at high pressures, leading to the discovery of a novel silicon-rich phase with enhanced superconducting properties. Through systematic crystal structure prediction, electronic structure calculations, and electron-phonon coupling analysis, we have identified a thermodynamically stable *I*₄₁/*amd*-YSi₆ phase that exhibits superconductivity with a critical temperature of 8.4 K at ambient pressure—a significant advancement over previously known binary yttrium silicides. The *I*₄₁/*amd*-YSi₆ phase features a unique 3D silicon network with distinctive dumbbell motifs, which creates an optimal structural framework for electron-phonon coupling. This structure remains dynamically stable across a wide pressure range (0–10 GPa), making it a promising candidate for experimental synthesis. Looking forward, experimental efforts to synthesize and characterize the predicted *I*₄₁/*amd*-YSi₆ phase will be crucial for validating our computational findings. High-pressure synthesis techniques followed by controlled decompression should be explored as a viable route to obtain this phase. Additionally, future computational and experimental investigations could focus on enhancing the *T_c* through strategic doping or alloying, exploring the effects of epitaxial strain on thin-film growth, and examining the potential for quantum coherence in devices based on these materials. This work not only expands our understanding of structure-property relationships in yttrium silicides but also provides a promising new candidate for silicon-based superconductors with potential applications in quantum computing and other advanced technologies.

Supporting Information

Supporting Information is available from the Wiley Online Library or from the author.

Acknowledgements

This work was supported by the National Science Foundation, Division of Materials Research (NSF-DMR) under Grant No. 2226700, and startup funds of the office of the Dean of SASN of Rutgers University-Newark. The authors acknowledge the Office of Advanced Research Computing (OARC) at Rutgers, The State University of New Jersey, for providing access to the Amarel cluster and associated research computing resources that have contributed to the results reported here.

Conflict of Interest

The authors declare no conflict of interest.

Data Availability Statement

The data that support the findings of this study are available in the supplementary material of this article.

Keywords

crystal structure prediction, rare-earth silicides, superconductivity

Received: May 14, 2025

Revised: June 16, 2025

Published online: July 9, 2025

- [1] W. Zhang, A. T. Bollinger, R. Li, K. Kisslinger, X. Tong, M. Liu, C. T. Black, *Sci. Rep.* **2021**, *11*, 2358.
- [2] Y. Chong, N. Hadacek, P. D. Dresselhaus, C. J. Burroughs, B. Baek, S. P. Benz, *Appl. Phys. Lett.* **2005**, *87*, 222511.
- [3] E. Bustarret, C. Marcenat, P. Achatz, J. Kačmarčík, F. Lévy, A. Huxley, L. Ortéga, E. Bourgeois, X. Blase, D. Débarre, J. Boulmer, *Nature* **2006**, *444*, 465.
- [4] L. J. Chen, *JOM* **2005**, *57*, 24.
- [5] H. F. Braun, *J. Less-Common Met.* **1984**, *100*, 105.
- [6] J. A. Flores-Livas, R. Debord, S. Botti, A. San Miguel, S. Pailhès, M. A. L. Marques, *Phys. Rev. B* **2011**, *84*, 184503.
- [7] H. Kawaji, H. Horie, S. Yamanaka, M. Ishikawa, *Phys. Rev. Lett.* **1995**, *74*, 1427.
- [8] S. Yamanaka, E. Enishi, H. Fukuoka, M. Yasukawa, *Inorg. Chem.* **2000**, *39*, 56.
- [9] D. Connétable, V. Timoshevskii, B. Masenelli, J. Beille, J. Marcus, B. Barbara, A. M. Saitta, G.-M. Rignanese, P. Mélinon, S. Yamanaka, X. Blase, *Phys. Rev. Lett.* **2003**, *91*, 247001.
- [10] M. Kriener, T. Muranaka, J. Kato, Z.-A. Ren, J. Akimitsu, Y. Maeno, *Sci. Technol. Adv. Mater.* **2008**, *9*, 044205.
- [11] E. A. Ekimov, V. A. Sidorov, E. D. Bauer, N. N. Mel'nik, N. J. Curro, J. D. Thompson, S. M. Stishov, *Nature* **2004**, *428*, 542.
- [12] P. Lejay, B. Chevalier, J. Etourneau, J. M. Tarascon, P. Hagenmuller, *Mater. Res. Bull.* **1983**, *18*, 67.
- [13] K. Kawashima, T. Muranaka, Y. Kousaka, S. Akutagawa, J. Akimitsu, *J. Phys.: Conf. Ser.* **2009**, *150*, 052106.
- [14] B. Liu, W. Yang, G. Xiao, Q. Zhu, S. Song, G.-H. Cao, Z. Ren, *Phys. Rev. Mater.* **2023**, *7*, 014805.
- [15] M. Imai, A. Sato, T. Aoyagi, T. Kimura, Y. Matsushita, N. Tsujii, *J. Am. Chem. Soc.* **2008**, *130*, 2886.
- [16] M. Imai, K. Nishida, T. Kimura, H. Abe, *Appl. Phys. Lett.* **2002**, *80*, 1019.
- [17] J. E. Baglin, F. M. d'Heurle, C. S. Petersson, *Appl. Phys. Lett.* **1980**, *36*, 594.
- [18] G. J. Campisi, A. J. Bevolo, F. A. Schmidt, *J. Appl. Phys.* **1981**, *52*, 6647.
- [19] J. A. Knapp, S. T. Picraux, *MRS Online Proc. Libr.* **1985**, *54*, 261.
- [20] W. Buckel, J. Wittig, *Phys. Lett.* **1965**, *17*, 187.
- [21] K. J. Chang, M. M. Dacorogna, M. L. Cohen, J. M. Mignot, G. Chouteau, G. Martinez, *Phys. Rev. Lett.* **1985**, *54*, 2375.
- [22] G. Martinez, J. M. Mignot, G. Chouteau, K. J. Chang, M. M. Dacorogna, M. L. Cohen, *Phys. Scr.* **1986**, *1986*, 226.
- [23] H. Fukuoka, J. Kiyoto, S. Yamanaka, *Inorg. Chem.* **2003**, *42*, 2933.
- [24] R. Vienneis, P. Toulemonde, C. Paulsen, A. San-Miguel, *J. Phys.: Condens. Matter* **2005**, *17*, L311.
- [25] P. Toulemonde, A. S. Miguel, A. Merlen, R. Vienneis, S. L. Floch, Ch. Adessi, X. Blase, J. L. Tholence, *J. Phys. Chem. Solids* **2006**, *67*, 1117.
- [26] J. Zhang, G. Chen, H. Liu, *J. Phys. Chem. Lett.* **2021**, *12*, 10388.
- [27] Y. Wang, J. Lv, L. Zhu, Y. Ma, *Phys. Rev. B* **2010**, *82*, 094116.
- [28] Y. Wang, J. Lv, L. Zhu, Y. Ma, *Comput. Phys. Commun.* **2012**, *183*, 2063.

- [29] S. Tao, X. Shao, L. Zhu, *J. Phys. Chem. Lett.* **2024**, 15, 3185.
- [30] G. Kresse, J. Furthmüller, *Phys. Rev. B* **1996**, 54, 11169.
- [31] J. P. Perdew, Y. Wang, *Phys. Rev. B* **1992**, 45, 13244.
- [32] J. P. Perdew, K. Burke, M. Ernzerhof, *Phys. Rev. Lett.* **1996**, 77, 3865.
- [33] P. E. Blöchl, *Phys. Rev. B* **1994**, 50, 17953.
- [34] H. J. Monkhorst, J. D. Pack, *Phys. Rev. B* **1976**, 13, 5188.
- [35] T. Rauch, M. A. L. Marques, S. Botti, *J. Chem. Theory Comput.* **2020**, 16, 2654.
- [36] T. Rauch, M. A. L. Marques, S. Botti, *Phys. Rev. B* **2020**, 101, 245163.
- [37] A. Togo, F. Oba, I. Tanaka, *Phys. Rev. B* **2008**, 78, 134106.
- [38] A. Togo, I. Tanaka, *Scr. Mater.* **2015**, 108, 1.
- [39] P. Giannozzi, S. Baroni, N. Bonini, M. Calandra, R. Car, C. Cavazzoni, D. Ceresoli, G. L. Chiarotti, M. Cococcioni, I. Dabo, A. D. Corso, S. de Gironcoli, S. Fabris, G. Fratesi, R. Gebauer, U. Gerstmann, C. Gougoussis, A. Kokalj, M. Lazzeri, L. Martin-Samos, N. Marzari, F. Mauri, R. Mazzarello, S. Paolini, A. Pasquarello, L. Paulatto, C. Sbraccia, S. Scandolo, G. Sclauzero, A. P. Seitsonen, et al., *J. Phys.: Condens. Matter* **2009**, 21, 395502.
- [40] G. Prandini, A. Marrazzo, I. E. Castelli, N. Mounet, N. Marzari, *npj Comput. Mater.* **2018**, 4, 1.
- [41] W. L. McMillan, *Phys. Rev.* **1968**, 167, 331.
- [42] P. B. Allen, R. C. Dynes, *Phys. Rev. B* **1975**, 12, 905.
- [43] W. Tang, E. Sanville, G. Henkelman, *J. Phys.: Condens. Matter* **2009**, 21, 084204.
- [44] U. Schwarz, A. Wosylus, H. Rosner, W. Schnelle, A. Ormeci, K. Meier, A. Baranov, M. Nicklas, S. Leipe, C. J. Müller, Y. Grin, *J. Am. Chem. Soc.* **2012**, 134, 13558.
- [45] S. Y. Savrasov, D. Y. Savrasov, *Phys. Rev. B* **1996**, 54, 16487.
- [46] F. Giustino, *Rev. Mod. Phys.* **2017**, 89, 015003.